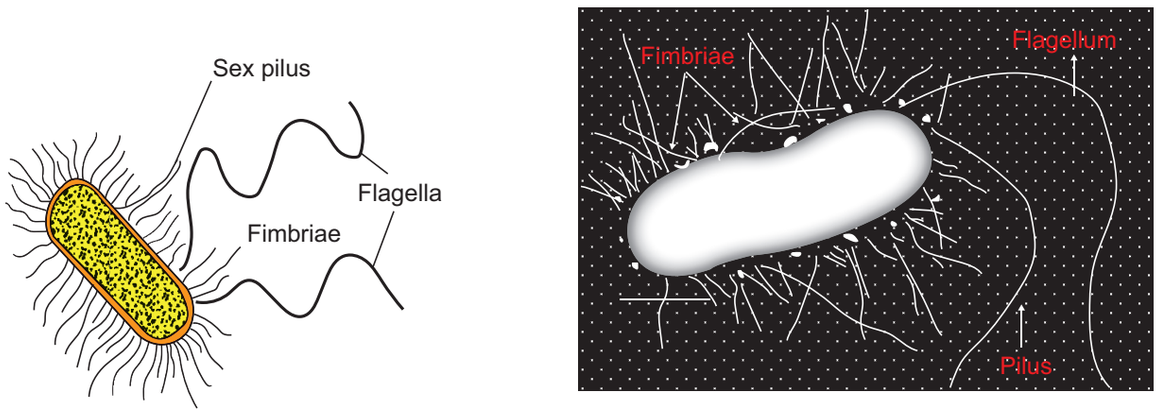


Figure 7.5: Structures of pili and fimbriae

Table 7.2 compares the pili and fimbriae.

7.2.2 Extracellular Polymeric Substance (EPS)

Many bacteria secrete high molecular weight polymers that adhere to the exterior of the cell wall to form a capsule or slime layer. Glycocalyx is often used to refer to any polysaccharide material outside the cell wall. Capsules and slime layer are considered to be glycocalyxes (Table 7.3).

Capsules

Some bacterial cells are surrounded by a viscous substance forming a covering layer or envelope around the cell wall called capsule (Figure 7.6). Capsule is usually made up of polysaccharide. It may be homopolysaccharide (made up of a single kind of sugar) or heteropolysaccharide (made up of several kinds of sugars). These are synthesized from sugars within the cell, transported and polymerized outside the cell. The capsule of some bacteria is made of polypeptides. The capsule of *Bacillus anthracis* has polymer of D-glutamic acid. Capsules are highly impermeable. Capsules can be demonstrated using special staining technique utilizing Indian ink or with Nigrosin stain. The presence of capsule in fresh isolates gives a moist and shiny appearance to the bacterial colonies on an agar medium. Capsular material is antigenic and may be demonstrated by serological methods.

Table 7.2: Comparison of pili and fimbriae

Characteristics	Pili	Fimbriae
Appearance	Hair like, straight appendages.	Tiny bristle like fibers arising from the surface of bacterial cell.
Length	Longer than fimbriae	Shorter than pili
Numbers per cell	1-10/cell	200-400/cell
Presence	Present only in Gram negative bacteria	Present in both Gram positive and Gram negative bacteria
Made -up of	Pilin protein	Fimbrillin protein

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